

	$\lambda_{\text{eff}} = 1.5$
7(a)	They are waves of the same frequency leading to a constant phase difference between them.
7(b)	$d \sin \theta = n\lambda$ $\left(\frac{0.001}{300}\right) \sin \theta = 2(640 \times 10^{-9}) \Rightarrow \theta \approx 22.6^\circ$ $\tan \theta = \frac{y}{2.1} \Rightarrow y = 2.1 \times \tan(22.6^\circ) \approx 0.873 \text{ m}$
7(c)	They are waves of the same frequency leading to a constant phase difference between them.